Chapter 1, Problem 26P

Problem

A cell phone battery is rated at 3.85 V and can store 10.78 watt-hours of energy.

Step-by-step solution

Step 1 of 3

(a)

A cell phone battery which is rated at 3.85 V, stores 10.78 watt-hours of energy.

The power absorbed by the battery over a period of 3 hours is,

$$\begin{aligned} p &= \frac{w}{t} \\ &= \frac{10.78}{3} \\ &= 3.59\text{ W} \end{aligned}$$

The average current it can deliver is,

$$\begin{aligned} i &= \frac{p}{v} \\ &= \frac{3.59}{3.85} \\ &= 0.93\text{ A} \end{aligned}$$

Therefore, average current delivered by the battery over a period of 3 hours is $\boxed{0.93\text{ A}}$.

Step 2 of 3

(b)

The average power delivered by the battery is,

$$\begin{aligned} p &= \frac{w}{t} \\ &= \frac{10.78}{3} \\ &= 3.59\text{ W} \end{aligned}$$

Therefore, the average power battery can deliver is $\boxed{3.59\text{ W}}$.

Step 3 of 3

(c)

The average current delivered by the battery is,

$$\begin{aligned} i &= \frac{p}{v} \\ &= \frac{3.59}{3.85} \\ &= 0.93\ \text{A} \end{aligned}$$

The ampere-hour rating of the battery is,

$$(3)(0.93) = 2.79\ \text{ampere-hour}$$

Therefore, the ampere-hour rating of the given cell phone battery is $\boxed{2.79}$.